



Building a Profitable Signals Model

The principles behind Alpha_X and Beta_X





Disclaimer

- This is not an investment proposal and/or financial advice. Educational purposes only.
- Any statement, even if said with absolute confidence, can be incorrect.
- The following information is based on our experience (we didn't conduct a scientific research).



Project_X

- Building models for tradable high return portfolios
- Prediction horizon: weeks, daily trading (though we are testing shorter term too)
- Fully automated stock picking and trading
- Using a spectrum of data points
- Utilizing ML algorithms
- Two models: Long and Long/Short (beta ~ 0)

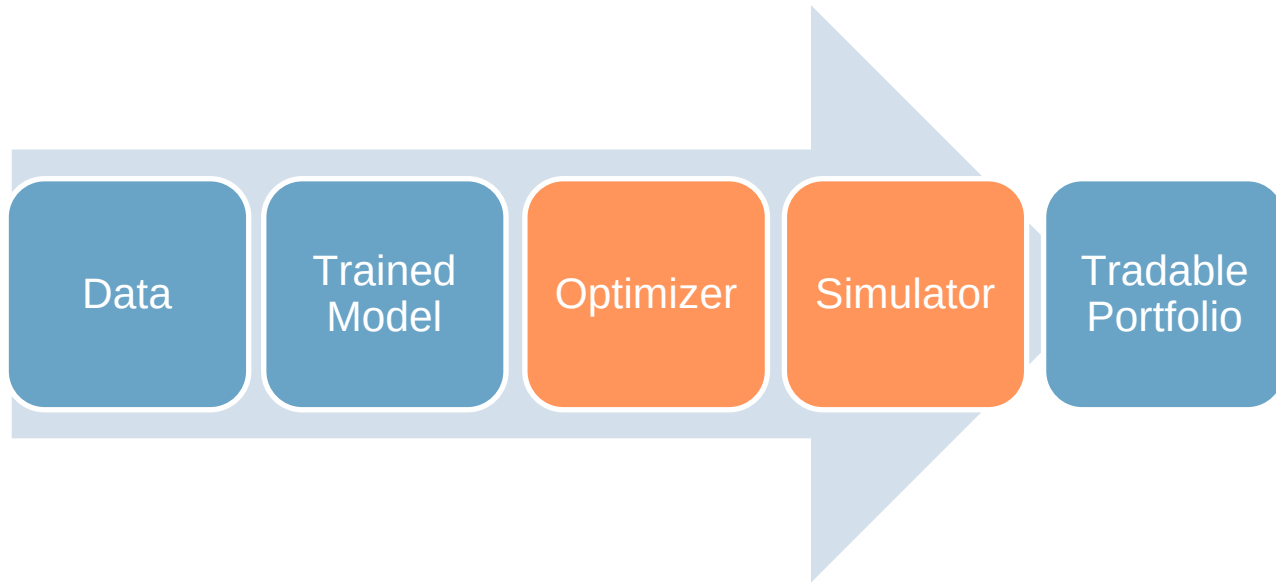


So, why Numerai's Signals?

- Another way for evaluating and validating our models
- Great, knowledgeable community
- Bonus: Potentially, generating additional stream of returns

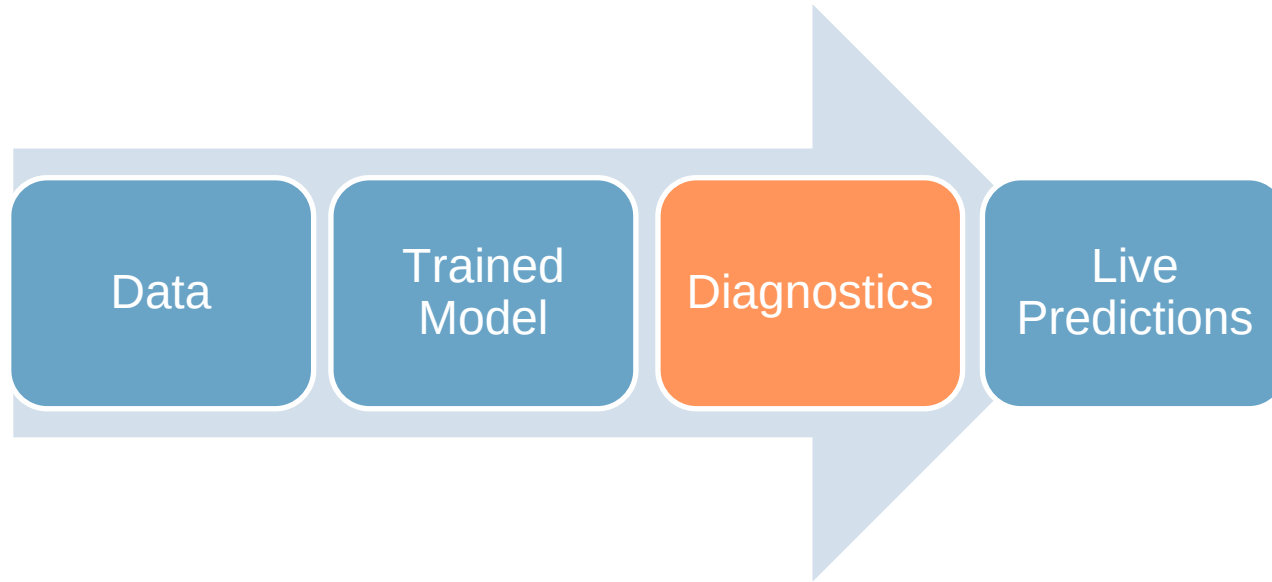


From Data to Tradable Model – Real Market



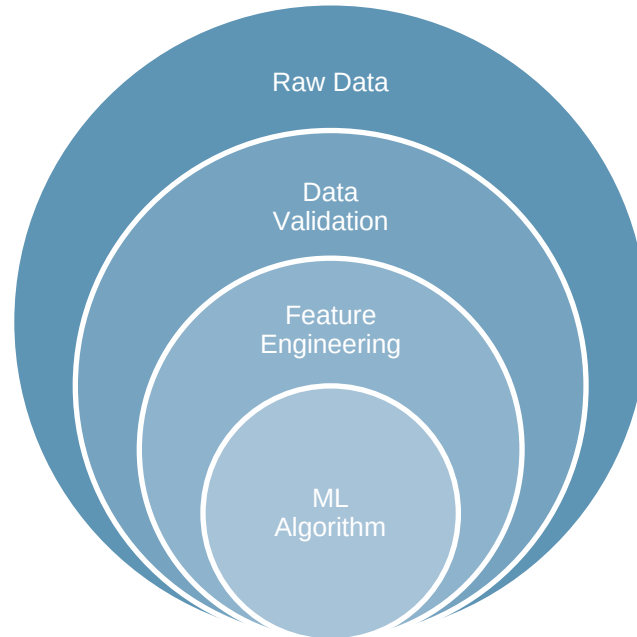


From Data to Signals Model





Components Importance





Data Types

- Fundamentals (Market Cap, Sector...)
- Pricing (OHLC, Volume, Technical Indicators)
- Quarterly Earning Reports
- News/Sentiments
- Insiders Activity (SEC reporting)
- Analyst Ratings
- Institutional Trade Activity (DTCC, 13F)
- Derivatives Activity (Options data)
- Alternative (google trends, satellite images, etc...)

Strong Signal – Follow the Hedge Funds (DTCC T+3)



Presented with permission from ExtractAlpha. For more information visit: <https://extractalpha.com/>





Data Cleaning

- Even the best data providers publish inaccurate data - validate it
- Different symbol naming conventions (Ticker, Bloomberg Ticker, ISIN etc...)
- Different stocks with same symbol
- Different symbols same stock (ex: FB and META)
- Stocks with trading suspension
- And more

Data Providers - Where to start



polygon.io



Alpha Vantage



* licensing is unknown

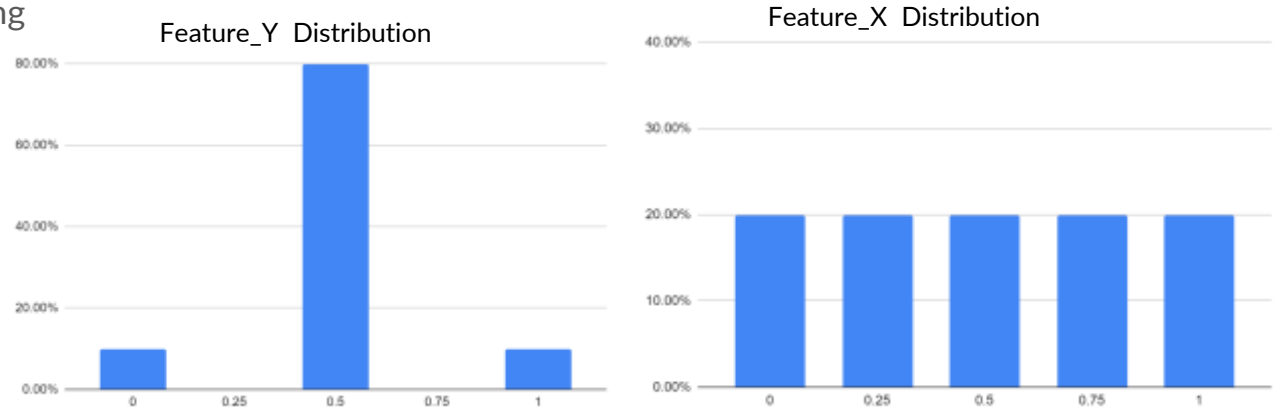


Feature Engineering - High Level Process

- Brainstorming
- Deciding
- Creating Features
- Testing
- Improving
- Repeat

Feature Engineering - Creating Features

- Numerical Transformation - scaling/normalization/binning/OHE
- Era/Week Based vs Cross Era
- Feature Extraction
- EDA/PCA/Clustering





Numerai Universe

- Predict on as many stocks as possible – significant improvement
- Update the universe weekly before submission – if you are binning the features, bin them after you filter out non-universe stock



Data Leakage

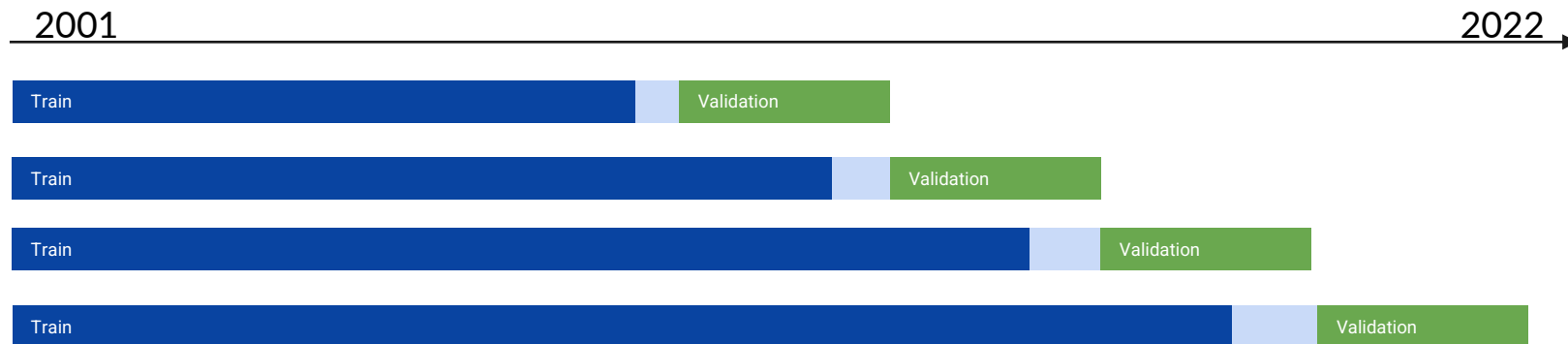
Training/Validating on information which would not be available in prediction mode

Examples:

- Earning Data: Quarter End Date vs Publish Date
- Non adjusted OHLCV
- Scaling/Normalization/Binning (non era-based)



Training / Validation - Time Series





Training / Validation - Time Series - Blocked CV





Targets

- Explore with several targets
 - Numerai Target
 - Return Target
 - Risk Adjusted Return Target
 - etc.
- Ensemble different models with different targets → improved stability (Sharpe) but can be a double edge sword

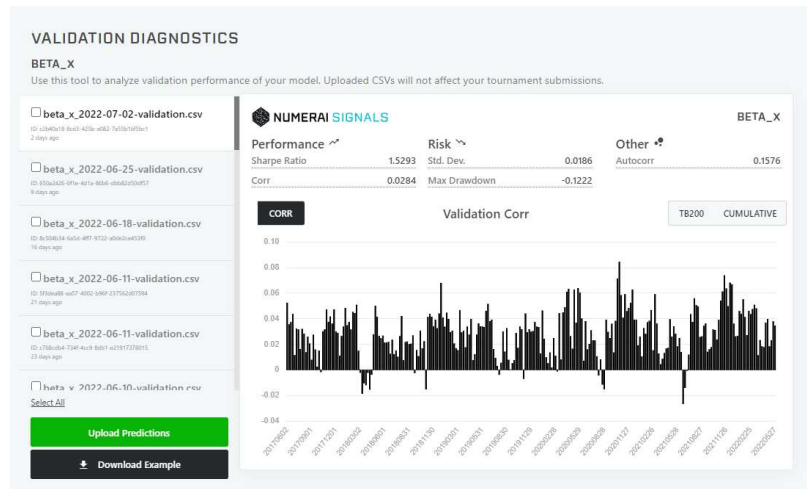


ML Algorithm

- GB Decision Trees: XGBOOST/LGBM
- Neural Networks
- Better **data/features** > model's hyper parameters

Diagnostics

- Your best tool for assessing the model performance before submitting
- Can indicate data leakage
- Can be used weekly to validate a proper weekly flow



VALIDATION DIAGNOSTICS

BETA_X

Use this tool to analyze validation performance of your model. Uploaded CSVs will not affect your tournament submissions.

☐ beta_x_2022-07-02-validation.csv

ID: e2b40a18-8cd3-425b-a082-7a55b16f5bc1

2 days ago

☐ beta_x_2022-06-25-validation.csv

ID: 650a1426-0f1e-4d1a-86b6-dbb62a50df57

9 days ago

☐ beta_x_2022-06-18-validation.csv

ID: 8c504b34-6a5d-4ff7-9722-a0de2ce453f0

16 days ago

☐ beta_x_2022-06-11-validation.csv

ID: 5f3dea88-aa57-4002-b96f-237562d07594

21 days ago

☐ beta_x_2022-06-11-validation.csv

ID: c768cbb4-734f-4cc9-8db1-e21917379d15

23 days ago

☐ beta_x_2022-06-10-validation.csv

[Select All](#)

Upload Predictions

Download Example



BETA_X

Performance

Sharpe Ratio

1.5293

Risk

Std. Dev.

0.0186

Other

Autocorr

0.1576

Corr

0.0284

Max Drawdown

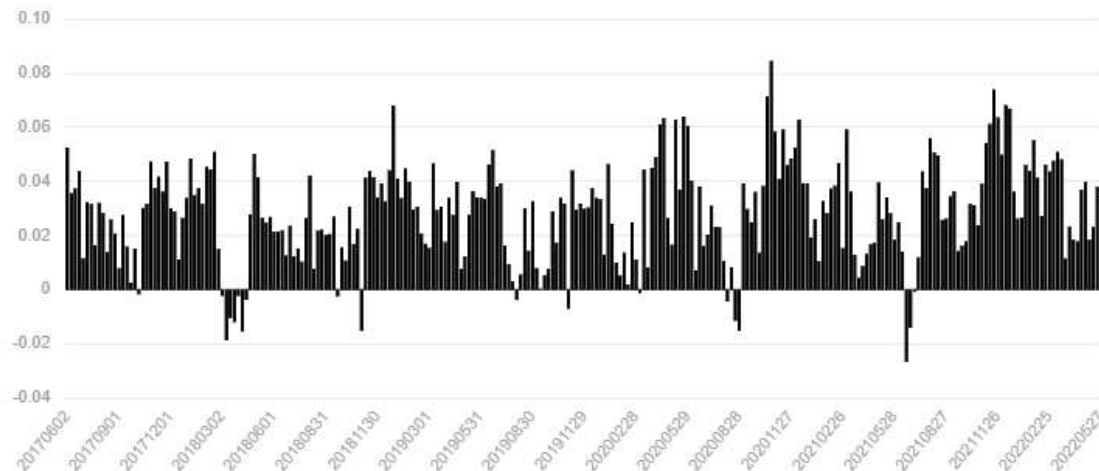
-0.1222

CORR

Validation Corr

TB200

CUMULATIVE



Live Performance

- The leaderboard provides limited information - overestimation danger
- Stability (Sharpe, Sortino) > Corr

→ 7	MM_T22	0.0200
→ 8	ALPHA_X	0.0200
9	ANNON_01	0.0105
→ 10	lg_t22_v3	0.0101
11	AZ_FEB_v3_Q27BQ2_INT_001R1	
12	KHTK49_S01	0.0187
→ 13	NN_T22_FULL	0.0187
→ 12	BETA_X	0.0184

	annual_return_2xcorr	annual_return_3xmmc	sharpe_2xcorr	sharpe_3xmmc	sortino_2xcorr	sortino_3xmmc
beta_x	1.865883	3.524448	1.756788	1.685113	2.256535	2.435640
lg_tx_fc1092_full	2.214423	4.389258	1.837376	1.888387	2.497610	2.417566
mm_t22	1.937118	3.755673	1.711651	1.671828	2.519157	2.218899
lg_t22_v3	1.960676	3.805010	1.691909	1.650417	2.353761	2.139898
alpha_x	2.141040	4.231654	1.699730	1.680044	2.205454	1.991244
lg_t44_v3	1.706442	3.288003	1.467147	1.370220	2.219146	1.909107
mm_t44	1.693995	3.293002	1.474378	1.380800	2.216739	1.879612
nn_t22_tr	1.545344	3.086224	1.397393	1.383521	1.998819	1.876343
xg_t44_tr	1.657572	3.002306	1.580218	1.387991	2.120044	1.759097

*Based on Degerhan Usluel notebooks

Improving Your Models

- Add new features (one by one? group?)
- For every new feature complete a full evaluation process
- Sources
 - Read financial news/blogs
 - Read markets prediction research





Staking

- Stake proportionally on uncorrelated stable models
- Balance periodically
- Take into consideration NMR and Numerai risks
- Set your goals and withdraw your profits accordingly
- TC is very speculative - be careful!

	Alpha_X	Beta_X	nn_tx_fc1092
Alpha_X	1.000	0.749	0.676
Beta_X	0.749	1.000	0.704
nn_tx_fc1092	0.676	0.704	1.000



Automate the Flow

- Update Universe
- Download Raw Data
- Build Full Dataset
- Predict both live and validation
- Submit Live / Test Validation (sanity)


Thank You

Rocketchat: @boogies999

